SPA617

RADIO ASTRONOMY LABORATORY

Experiment : Antenna Characterization

Name : Aditya Waghmare

Roll No. : 261310001

Programme : M.Tech

Antenna Characterization

Aim : To determine the antenna operating frequency and return loss from **VSWR** measurements.

Objective :

The remained part a) in Antenna Characterization Experiment which is " To determine the antenna operating frequency and return loss from VSWR measurements" is covered in this segment of Experiment .

Theory

Basic antenna concepts

An antenna is a transducer that converts electrical power into electromagnetic waves and vice versa. An antenna can be used either as a transmitting antenna or as a receiving antenna.

A dipole antenna has two equal metal arms in a straight line and is fed at the centre. Its total length is approximately half a wavelength ($\lambda/2$), with a feed line connected between the two arms, as shown in Fig. 1. It radiates most strongly in directions perpendicular to the wire and has very little radiation along the wire.

A monopole antenna is approximately half of a dipole mounted above a conducting ground plane. Its length is approximately one-quarter wavelength ($\lambda/4$). The ground acts like a mirror, making it behave like a full dipole.

- A transmitting antenna converts electrical signals into electromagnetic waves and radiates them.
- A receiving antenna converts the received electromagnetic waves into electrical signals.
- In two-way communication, the same antenna can be used for both transmission and reception.

Antenna beam pattern

The energy radiated by an antenna is represented by the radiation pattern of the antenna. The radiation pattern generally has a main lobe, side lobes, and a back lobe.

- The main lobe is the region where the maximum radiated energy exists. The direction of this lobe indicates the principal direction of the antenna beam.
- The other parts of the pattern where radiation is distributed sideways are known as side lobes or minor lobes.
- The lobe opposite to the main lobe is known as the back lobe.

E-plane and H-plane angular measurements are used to determine the radiation beam pattern of an antenna. The E-plane is the plane containing the electric-field vector and the propagation vector, while the H-plane is the plane containing the magnetic-field vector and the propagation vector. The E-plane and H-plane are perpendicular to each other.

Reflection coefficient, VSWR, and return loss

When the antenna impedance Z_L differs from the characteristic impedance Z_0 of the transmission line, a part of the incident signal is reflected. The voltage reflection coefficient at the antenna terminals is

$$\Gamma = \frac{V_{\text{reflected}}}{V_{\text{incident}}} = \frac{Z_L - Z_0}{Z_L + Z_0}.$$

The magnitude $|\Gamma|$ indicates the degree of impedance mismatch. For a perfectly matched antenna,

$$Z_L = Z_0, \qquad |\Gamma| = 0.$$

The voltage standing-wave ratio is related to the magnitude of the reflection coefficient by

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}.$$

The return loss is defined as

$$RL = -20\log_{10}|\Gamma| \qquad \text{dB}.$$

A well-matched antenna has a VSWR close to unity, a small reflection coefficient, and a large return-loss value. The resonant or operating frequency is generally identified near the minimum of the VSWR curve.

Components Required

- Monopole and dipole antennas
- RF signal generator
- Digital storage oscilloscope (DSO)
- Spectrum analyser
- RF Tee connector
- 50 Ω termination
- Antenna support stand
- Coaxial cables and suitable RF adapters

1.7 Experimental Procedures

1.7.1 Part A: VSWR , Return Loss and antenna operating frequency*

1. Connect the signal generator, RF Tee connector, antenna (DUT), and oscilloscope as given in Fig 5.

2. Set the oscilloscope input impedance to 1 MΩ.

3. Keep the signal-generator amplitude constant and select test frequencies across the ex pected operating range of the antenna.

4. Around each selected test frequency fk, perform a small-span frequency sweep contain ing one complete amplitude-ripple period.

5. During the sweep, record:

- Vmax: the maximum voltage amplitude observed on the oscilloscope;

• Vmin: the minimum voltage amplitude observed on the oscilloscope. Use the same voltage quantity for both measurements.

For example, if Vmax is measured as a peak-to-peak voltage, measure Vmin as a peak-to-peak voltage also.

6. Calculate the measured voltage standing-wave ratio: VSWR= Vmax / Vmin .

7. Calculate the magnitude of the reflection coefficient:

$|\Gamma| = (VSWR-1) / (VSWR+1) = (Vmax - Vmin) / (Vmax + Vmin)$.

8. Calculate the return loss: $RL=-20\log10|\Gamma|$ dB.

9. Repeat the measurement at different frequencies across the expected operating range of the antenna.

10. Plot VSWR and return loss versus frequency.

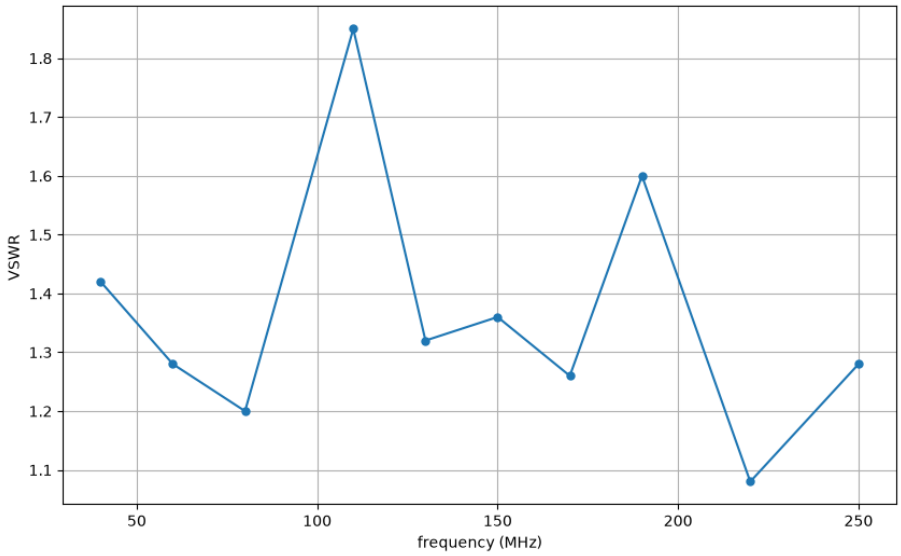
11. Identify the frequency at which the VSWR is minimum and state it as the approximate antenna operating or resonant frequency.

12. Determine the frequency range over which VSWR ≤ 2.

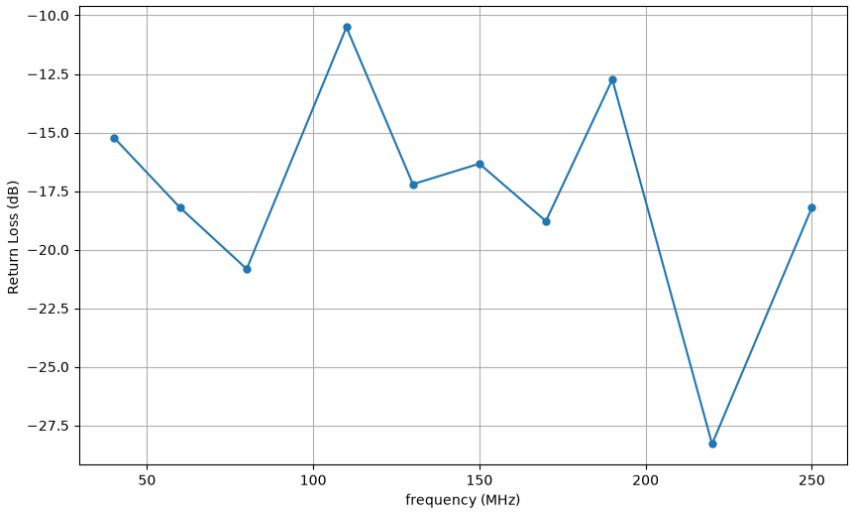
Observations and Results :

Frequency (MHz)	Vmax (mV)	Vmin (mV)	VSWR	Return Loss(dB)
40	485	340	1.42	-15.21
60	485	377	1.28	-18.21
80	456	377	1.20	-20.82
110	456	246	1.85	-10.50
130	327	246	1.32	-17.20
150	327	240	1.36	-16.33
170	304	240	1.26	-18.78
190	304	190	1.6	-12.73
220	207	190	1.08	-28.29
250	207	161	1.28	-18.21

Plot of Frequency (MHz) vs VSWR



Plot of Frequency (MHz) vs Return Loss (dB)



VSWR is less than 2 over the frequency range from 40 MHz to 250 MHz.

Compare the measured operating frequency with the theoretical resonance frequency of a $\lambda/4$ monopole or a $\lambda/2$ dipole, as applicable.

$\lambda/4 = 0.05\text{cm} \rightarrow \lambda = 0.2\text{m}$

$f_{operating} = c/\lambda = 1.5\text{GHz}$

To ensure proper beam pattern measurements , the minimum far-field distance was verified using the maximum aperture dimension (D=0.29m) and the signal generator power set to 20dBm :

$R_{far\text{-}field} = 0.841\text{m}$

Thus, all angular measurements are taken beyond this 84.1cm boundary.

Conclusions :

State the following results and briefly discuss their physical significance:

- measured antenna operating frequency, minimum VSWR, and return loss;

  The theoretical operating frequency for the 5 cm quarter-wave monopole was calculated 1.5GHz. The verified far-field distance is 841 cm boundary , so all angular measurements are taken beyond this 84.1cm boundary. The minimum VSWR was found at frequency 220 MHz which is 1.08 . Minimum VSWR and maximum return loss indicate good impedance matching, but they do not guarantee that an antenna radiates efficiently. An antenna may absorb power in resistive losses instead of radiating it.